Setting Up an Amazon RDS MySQL Database and Connecting To It

To Begin with the Lab

Summary of the Lab

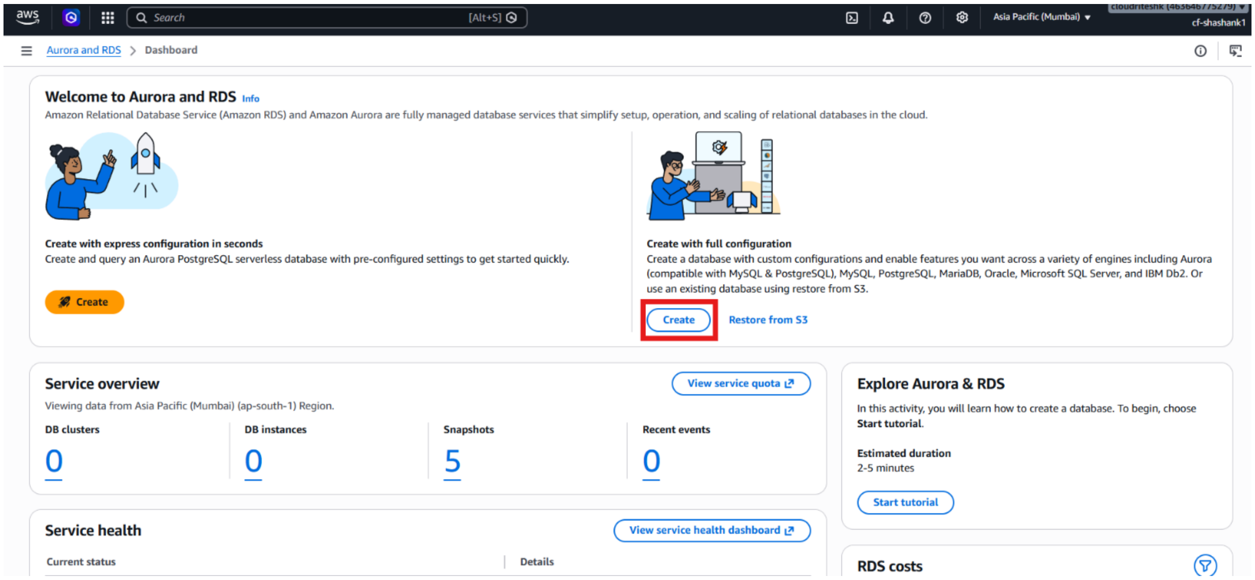
In this lab, an **Amazon RDS** MySQL database was created using the AWS Management Console with a minimal free-tier setup. The database engine version was selected, a database name, admin username, and password were configured, and public access was enabled so the database could be reached from a local SQL client. After the instance finished provisioning, the endpoint was copied and used in **MySQL Workbench** to establish a connection. A new database was then created from the client side to show how database administration is handled after the RDS instance is available. Finally, the setup was verified by successfully connecting to the database and confirming that the instance could be managed from the SQL client.

Understanding Amazon RDS and SQL Client Access

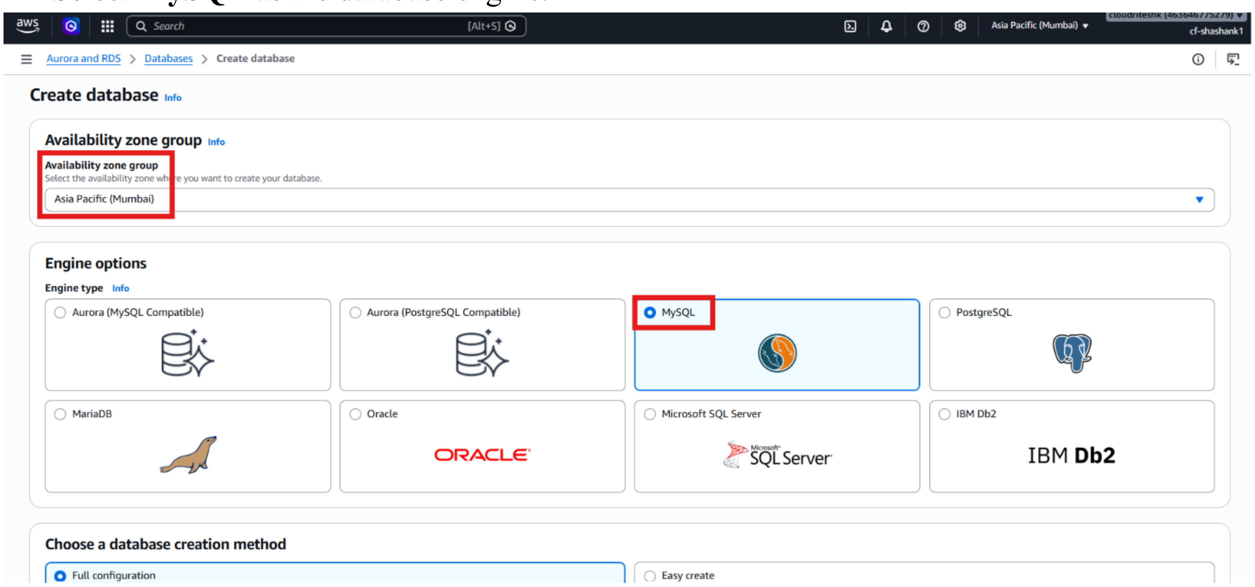
Amazon RDS is a managed database service that lets you create and run databases without handling the low-level server setup yourself. A SQL client such as **MySQL Workbench** is used to connect to the database, run SQL commands, and manage database objects like schemas and tables. This exists because cloud engineers usually create the database infrastructure, while database users or programmers handle the actual data design and SQL operations. Public access is convenient for lab practice, but in real projects the database is usually placed in a private network and accessed through a controlled application server or bastion host.
Example:
A cloud engineer creates a MySQL RDS instance named mydb with username admin and gives the endpoint to a database developer. The developer opens **MySQL Workbench**, connects using the RDS endpoint, and creates a new application database called myappdb. This separates infrastructure setup from database usage and makes administration easier.

Lab Steps

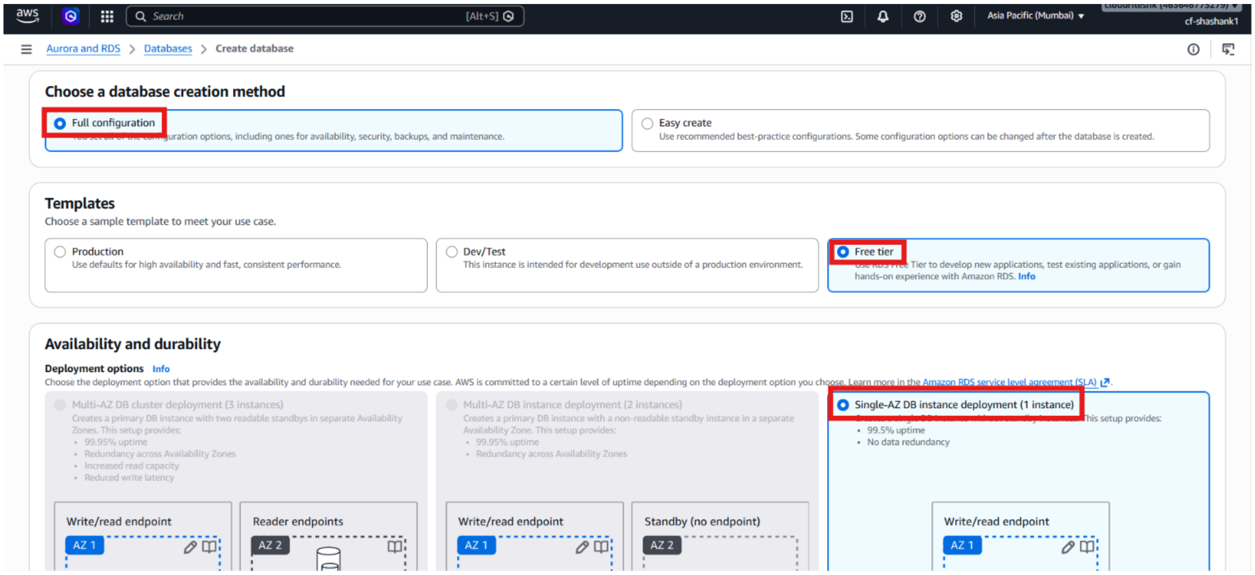
- Sign in to the AWS Management Console.
- Open **Aurora and RDS**.
- Choose **Create with full configuration** and Click **Create database**.



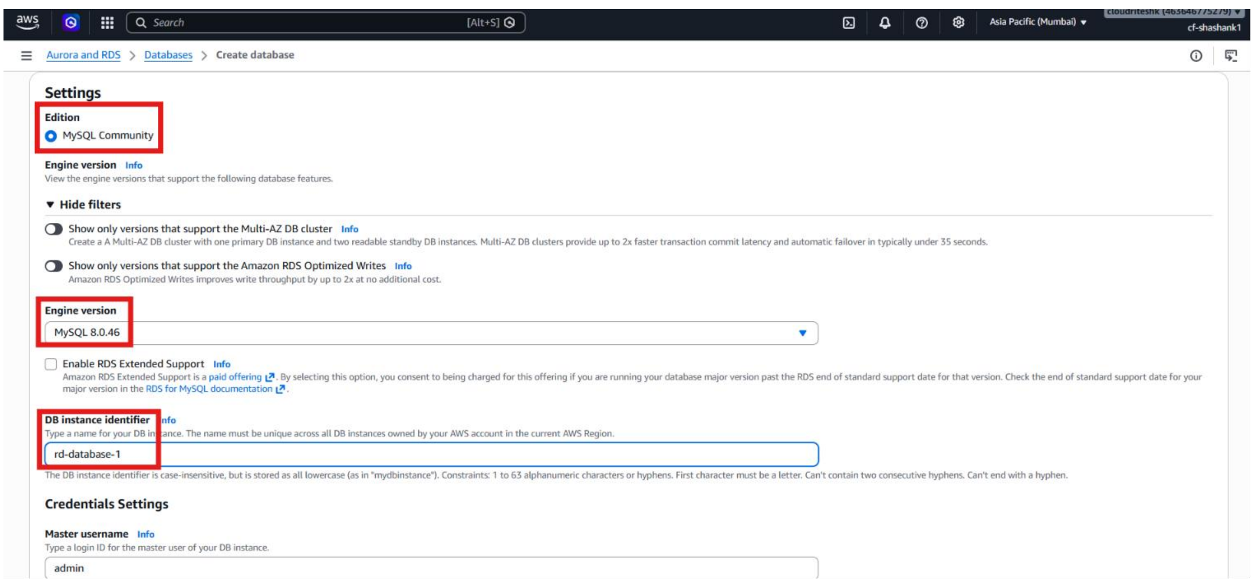
- Choose your **Availability Zone.**
- Select **MySQL** as the database engine.



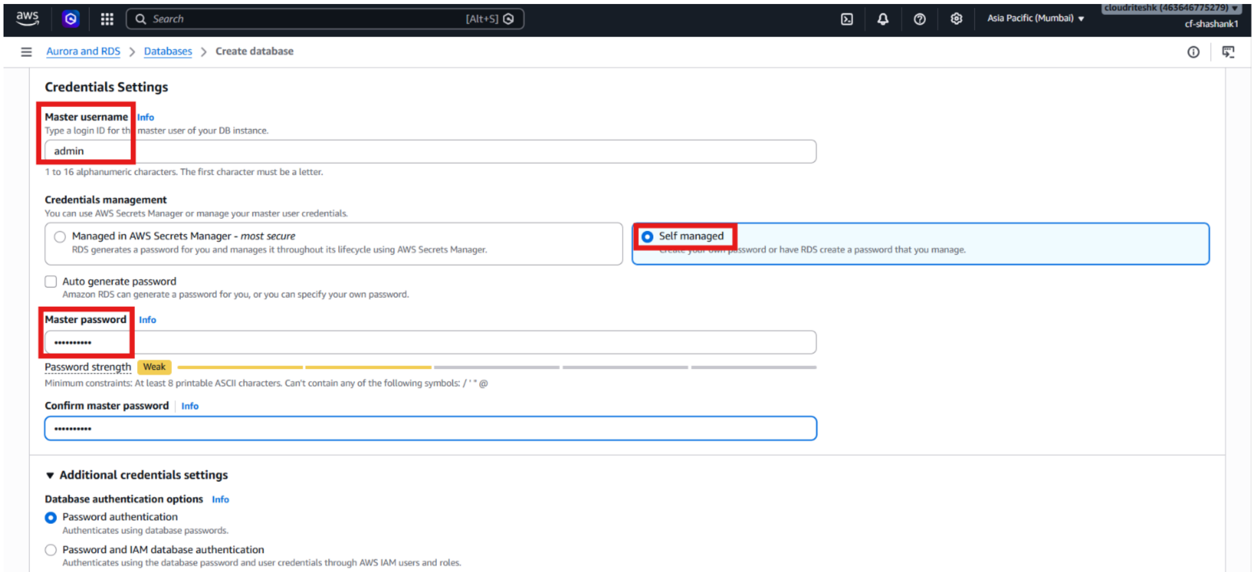
- Choose database creation method as **Full configuration.**
- Choose the **Free tier** template.



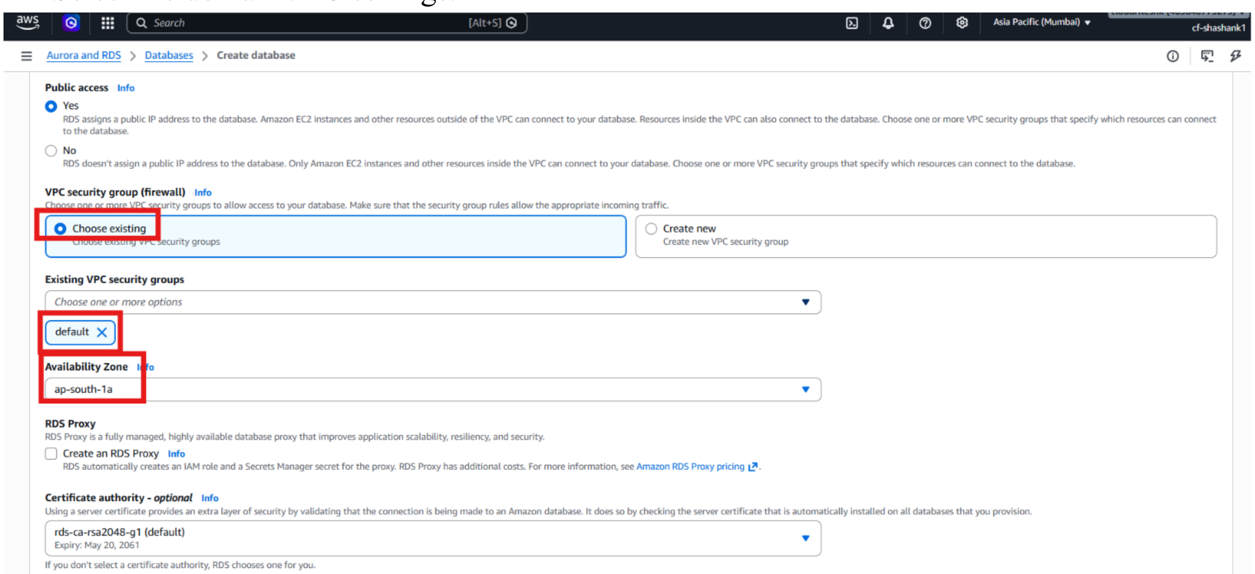
- Enter the database name.
- Keep the engine version at the default MySQL 8.0.x option or choose one you have installed on your computer.
- Set the master username to **admin**.



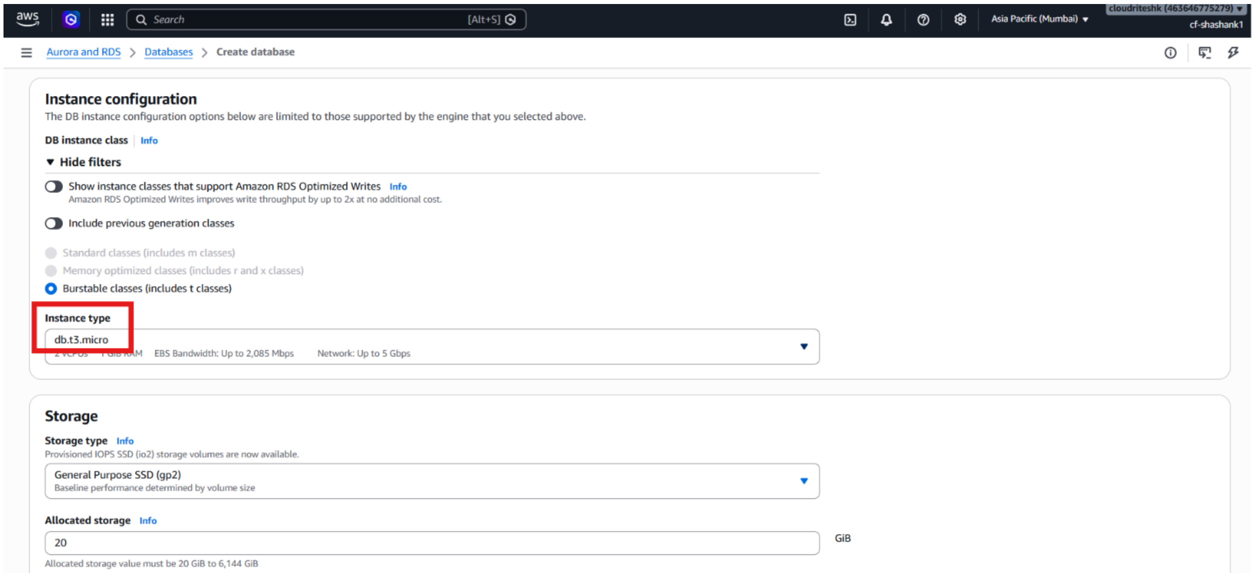
- Credentials manager as **Self-Managed.**
- Create and confirm a password.



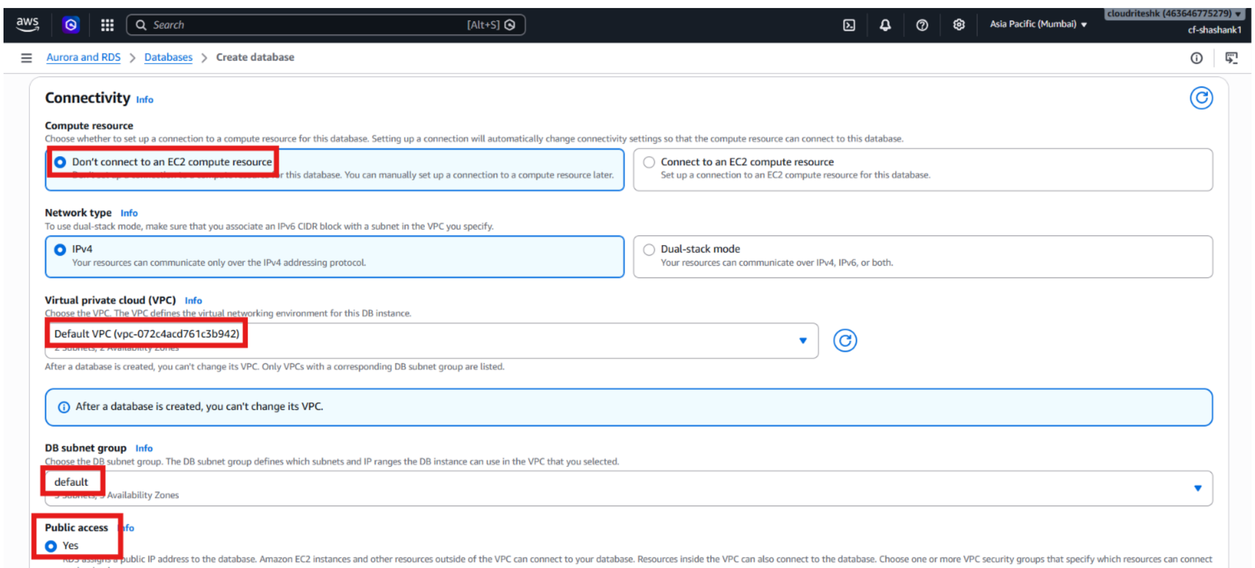
- Leave storage at the default value.
- Select the default VPC settings.



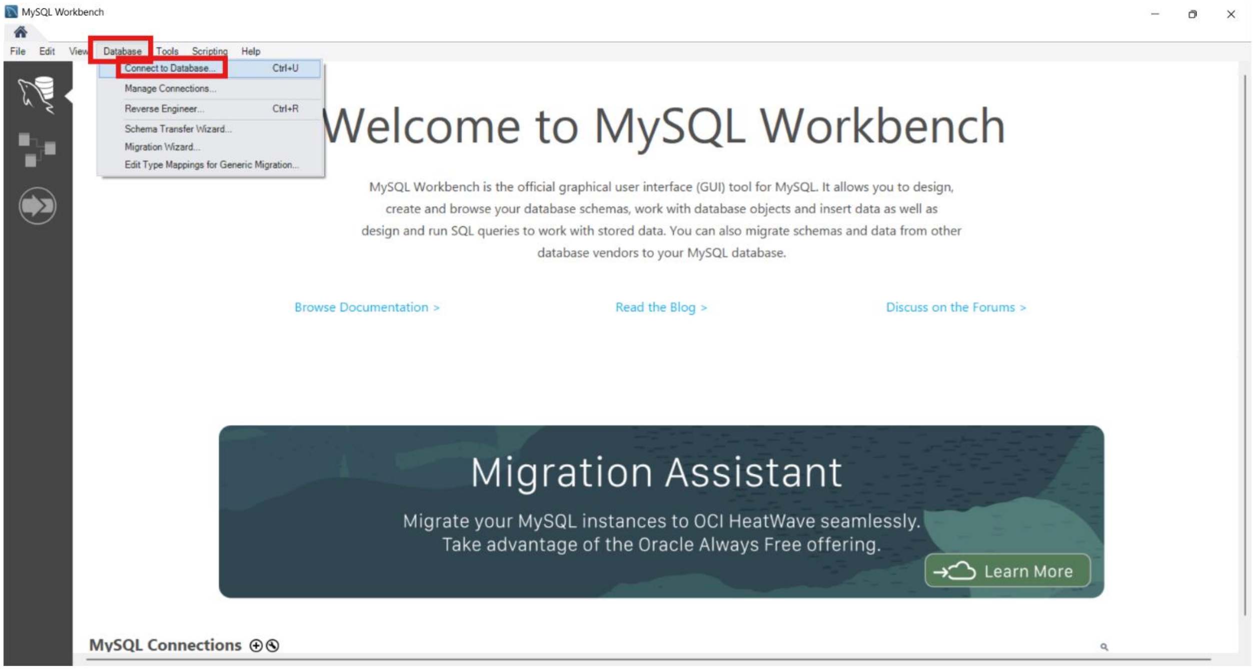
- Keep the default instance class provided by the free-tier setup.



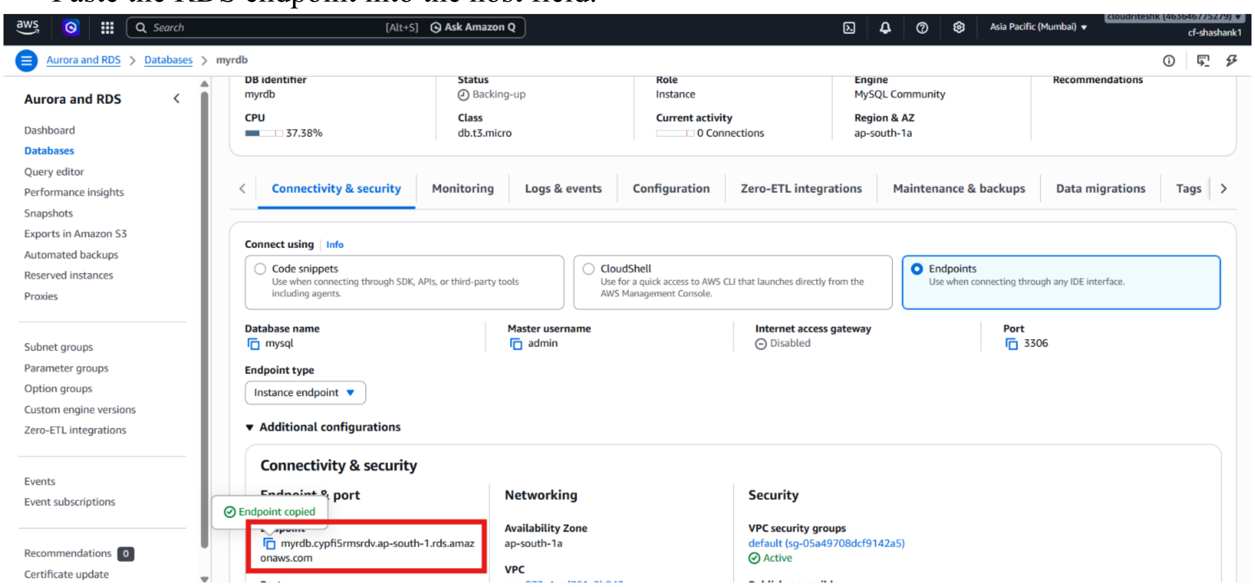
- Enable public access so the database can be reached from the local machine.
- Keep the default security group selection.
- Make sure to enable **Public Access.**



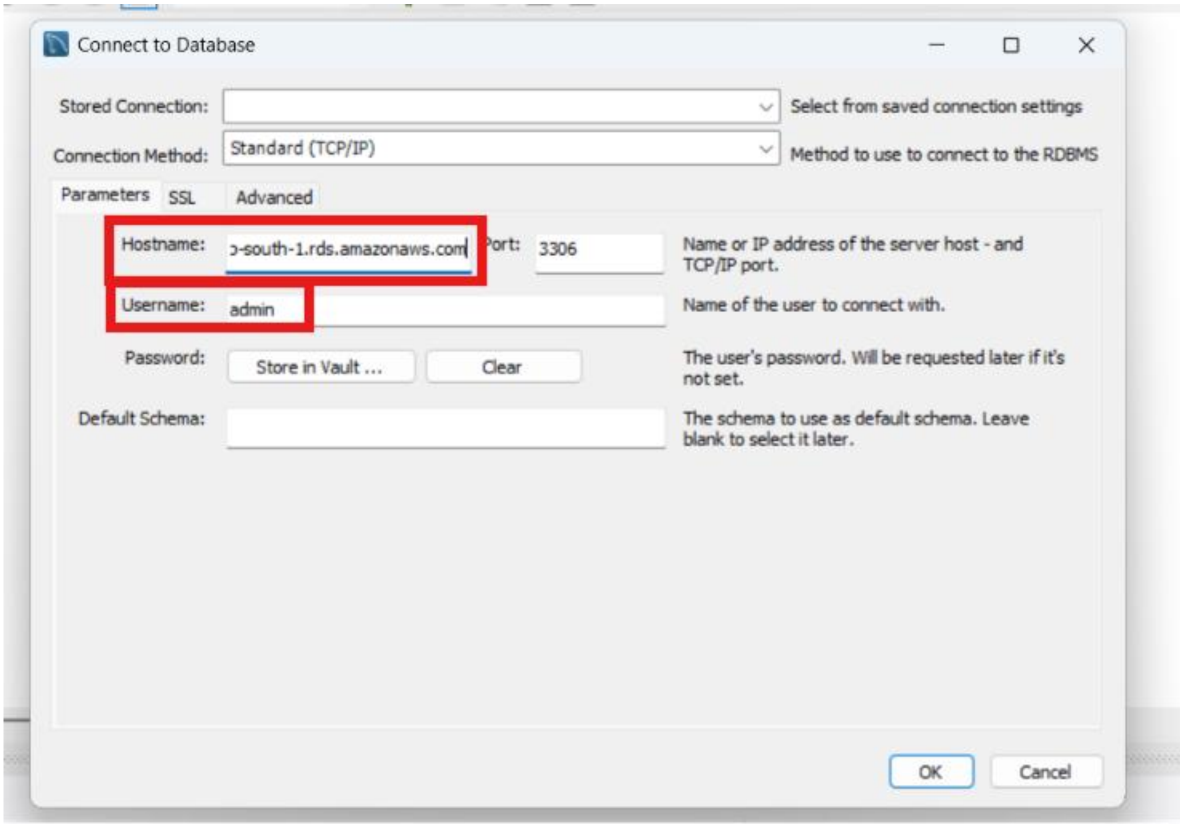
- Click **Create database**.
- Wait until the database status becomes **Available**.
- Copy the RDS endpoint from the database details page.
- Open **MySQL Workbench** on the local computer.
- Create a new MySQL connection.



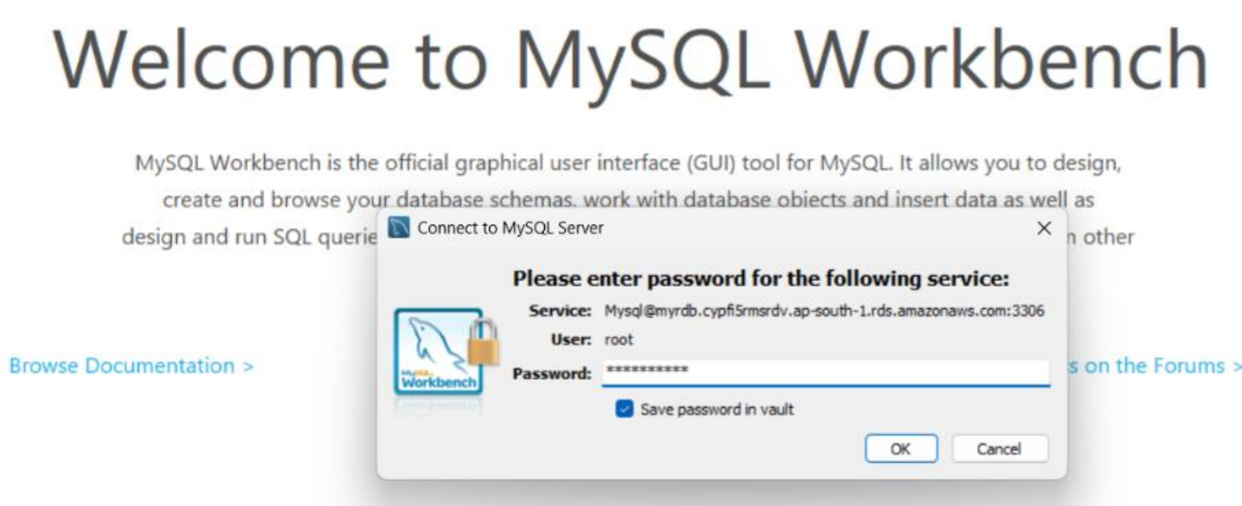
- Paste the RDS endpoint into the host field.



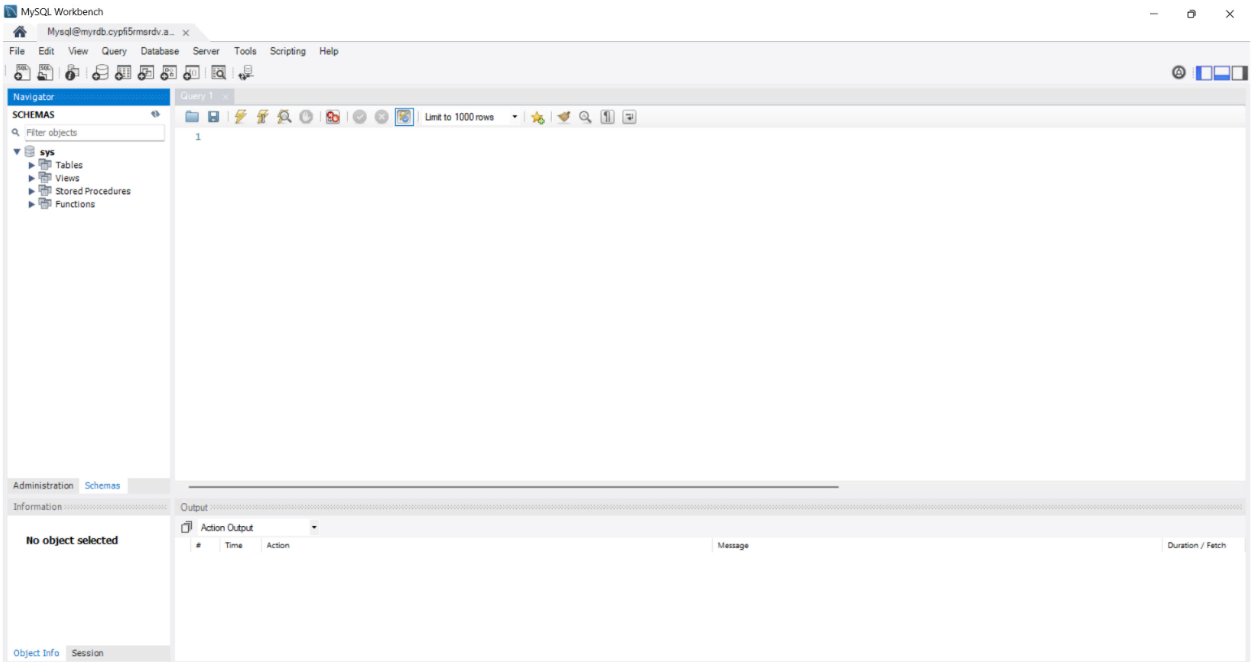
- Use **admin** as the username.
- Test the connection.



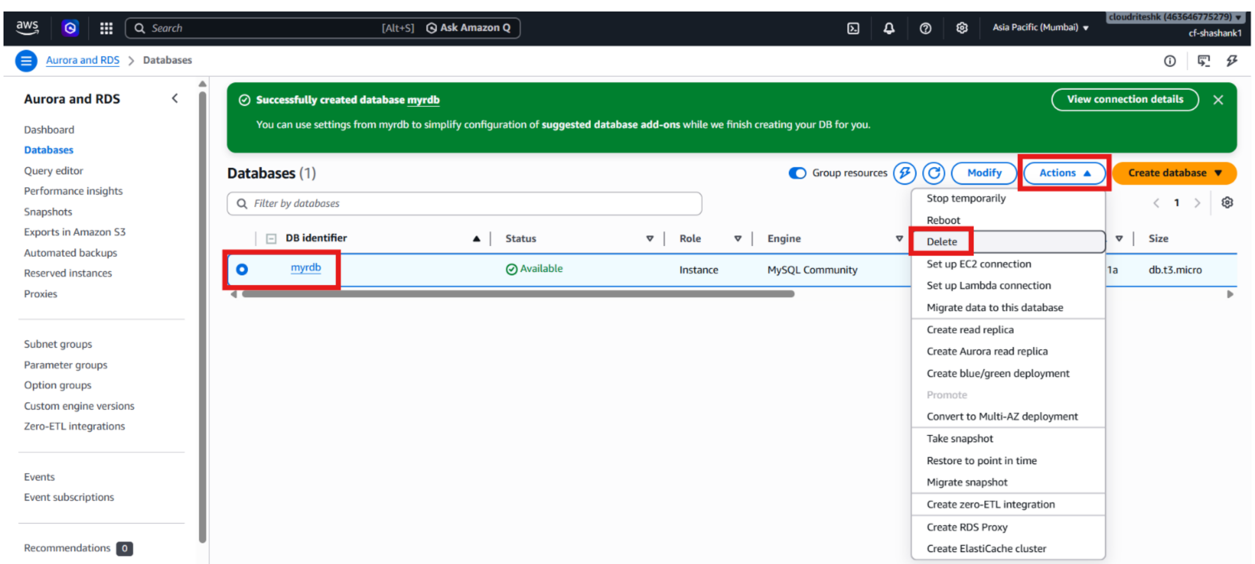
- Enter the password when prompted.
- Save the password if needed.



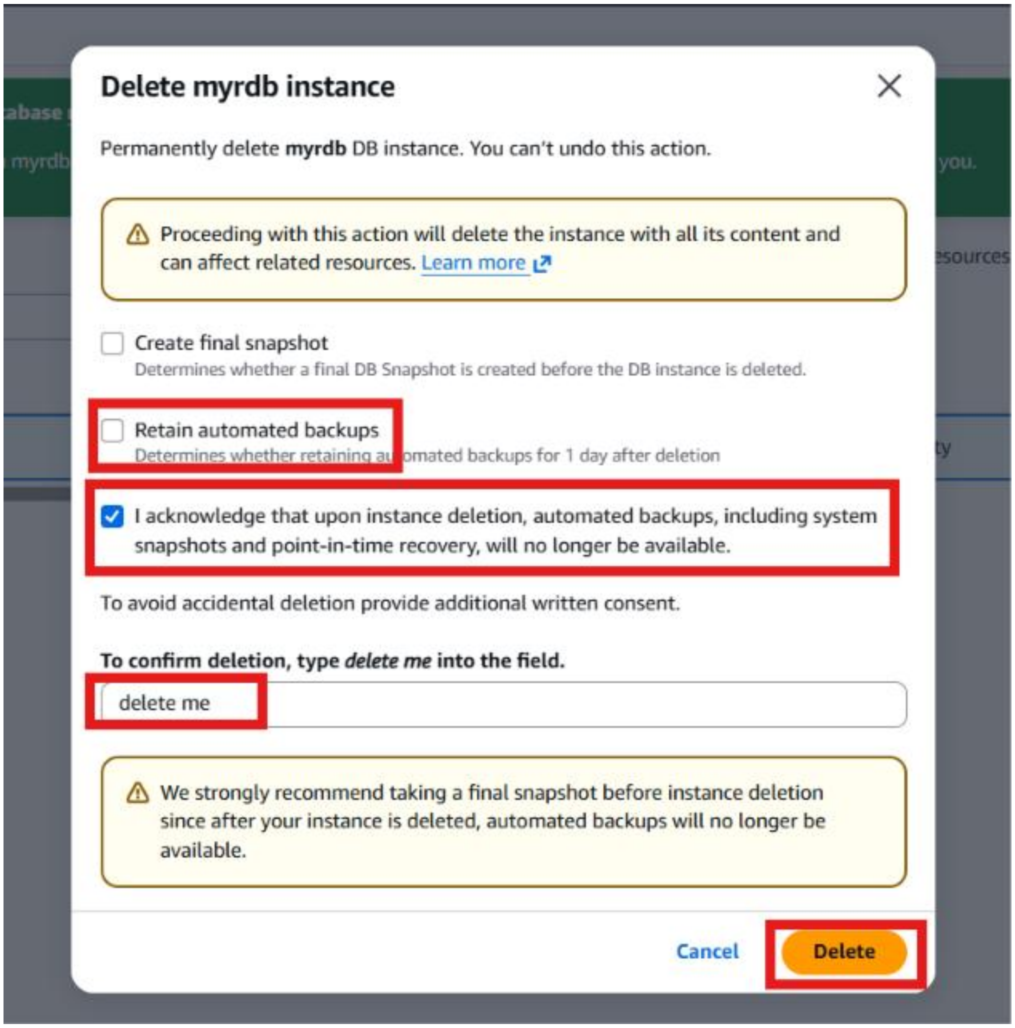
- Open the database connection successfully.
- Create a new database from **MySQL Workbench**.
- Verify that the new database appears in the schema list.



- Understand that database creation in the client is separate from RDS instance creation.
- Note that tables and schemas can now be created inside the new database.
- Return to the AWS console after testing the connection.
- Select the RDS database instance.
- Open **Actions**.
- Choose **Delete** to remove the lab database when finished.



- Type **delete me** when prompted.
- Confirm that automatic backup is disabled if required by the deletion flow.



- Finalize the deletion request.

Verification

- The **Amazon RDS** MySQL instance was created successfully.
- The instance reached the **Available** state.
- **MySQL Workbench** connected to the database using the RDS endpoint.
- A new database was created from the SQL client.
- The lab database was ready for cleanup after the demonstration.